LiDAR

LiDAR

- LiDAR is an acronym derived from *Light Detection and Ranging*
- *Also known* as LADAR (*LASER Detection and Ranging), and also* as LASER Altimetry
- LiDARs are based on an application of lasers, they use light that is composed of a very narrow band of wavelengths
- Such light can be transmitted over large distances as a narrow beam that will diverge only slightly, in contrast with most light in our everyday experience

LiDAR

- LiDAR is an active sensor
- LiDAR uses electromagnetic energy in the visible and near infrared wavelengths
- LiDAR is a nadir-looking instrument
- As LiDAR operates in the visible and near-infrared wavelengths, its signal is affected by atmospheric conditions
- LiDAR records information at discrete points across the swath

LiDAR

- LiDAR instrument measures the time taken by an energy pulse to reach the ground, and for a part of the scattered radiation to return to the sensor
- Higher the pulse rate more the swath/coverage but weaker the returns,

  (few hundred to about 30,000 per second)
- A single pulse may return multiple returns

LiDAR

- Rapid growth in the application of LiDAR are due to two technological developments
  - Improved electro-mechanical technology, sensors and lasers can be accurately and reliably controlled in moving aircrafts
  - Precise aircraft positioning and orientation using GPS , Inertial navigation systems (INS)
- Current attainable accuracy up to 15 cm
- DEM/DSM at a grid spacing of 1 meter

Important Components

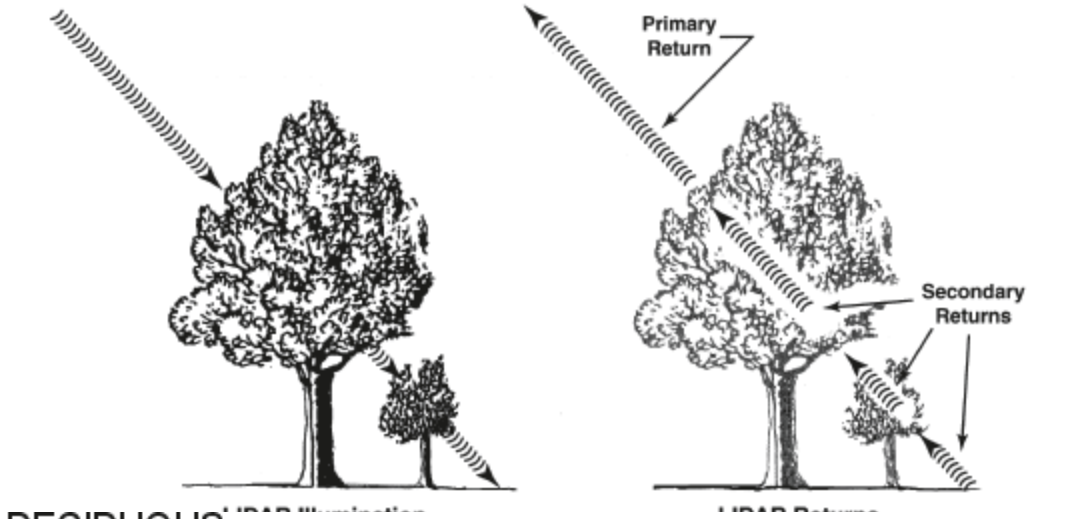
- **Inertial measurement units (IMUs):** enable precise control and recording of orientation of aircraft (roll, pitch, and yaw)
- **GPS**: provides accurate records of geographic location of an aircraft as it acquires data
- **Highly accurate clocks**: permits precise timing of LiDAR pulses required to create high performance LiDAR scanning systems

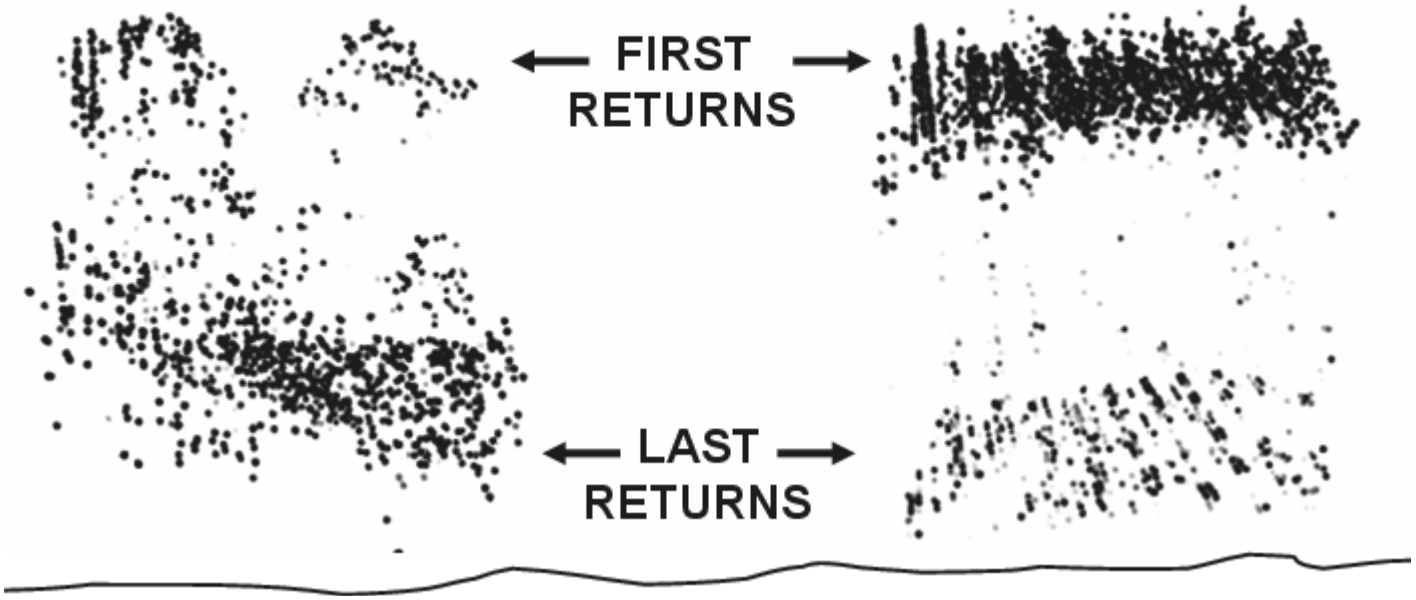
LiDAR

- LiDAR measures the distance from the sensor to the ground
- As the position of the instrument is known, accurate position of the each return pulse can be calculated, and the distances can be converted to elevations above a specified geodetic datum
- Raster map of these elevations can be generated by interpolation
- Set of produced elevation values does not necessarily define a digital elevation model (DEM), because the LiDAR pulse is reflected back by the first object of sufficient size and density that it meets as it travels downwards from the instrument
- These objects may be a branch, a tree crown, the ground surface, or the top of a building, depending on the area being viewed

LiDAR

- Term 'first return' or 'first bounce' system records the time taken between the emission of the pulse of light energy and the reception at the sensor of the first return
- while the second is the position of the last indication of backscatter. This second event is called the last return
- There is no difference between the first and last return if the target object does not transmit light
- if it does, then the time difference between the first and last bounce is proportional to the height of the object above the ground


Primary
Return
Secondary
Returns
DECIDUOUS
CONIFEROUS


FIRST
RETURNS
LAST
RETURNS

LiDAR DATA



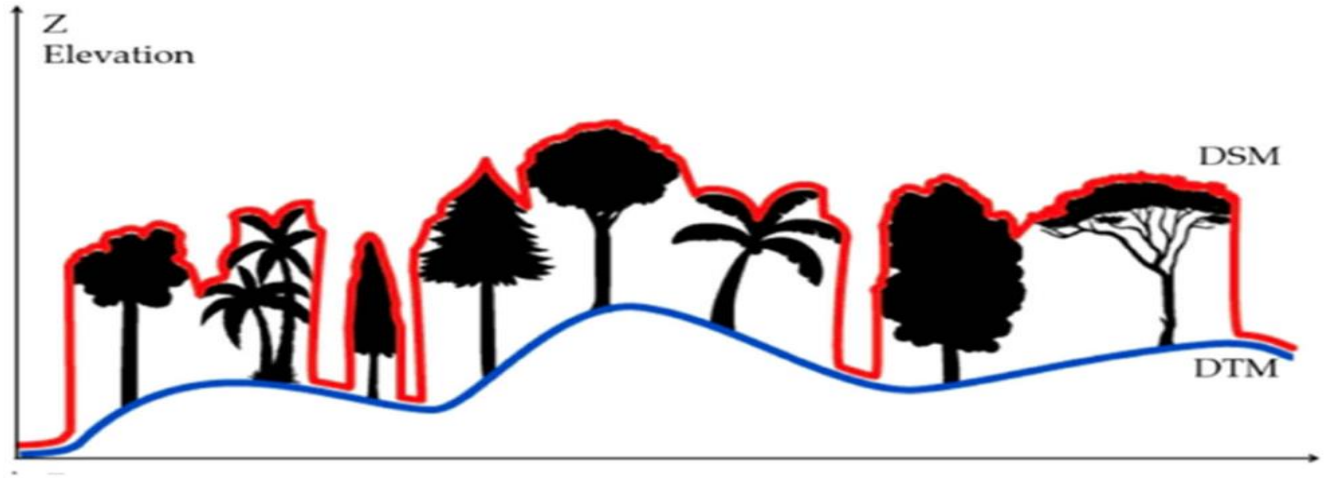
Data with brightness values



Processed data showing terrain

LiDAR

- Set of point elevation values can be used to generate a digital surface model (DSM), which shows the elevation of the highest reflective object on the ground
- LiDAR data can be used to map the highest point on a building, or of a tree
- To generate a DEM from the DSM, the surface objects must be removed, or the height of the superimposed object must be estimated
- DSM is useful in itself
- LiDAR images can be used to generate 3D urban models





Digital Surface Model Digital Terrain Model Canopy Height Model

Digital Elevation Model

- One of the common photogrammetric product
- Have increased number of utilization
- Represents the Earth's surface elevation digitally as an array of points
- DEM:
  - Elevation is indicated by the image intensity with higher point having brighter values

Digital Terrain Elevation Model

- Is also referred to as Digital Terrain Model (DTM) or Digital Terrain Elevation Model (DTEM)
- DSM (Digital Surface Model) and nDSM (normalized Digital Surface Model) –other forms of elevation representation
- DTM or DTEM- refers to the elevation of bare terrain/Earth surface
- DSM- Representation of elevation of elevated objects such as buildings and trees
- DEM- Representation of elevation

Real Situation



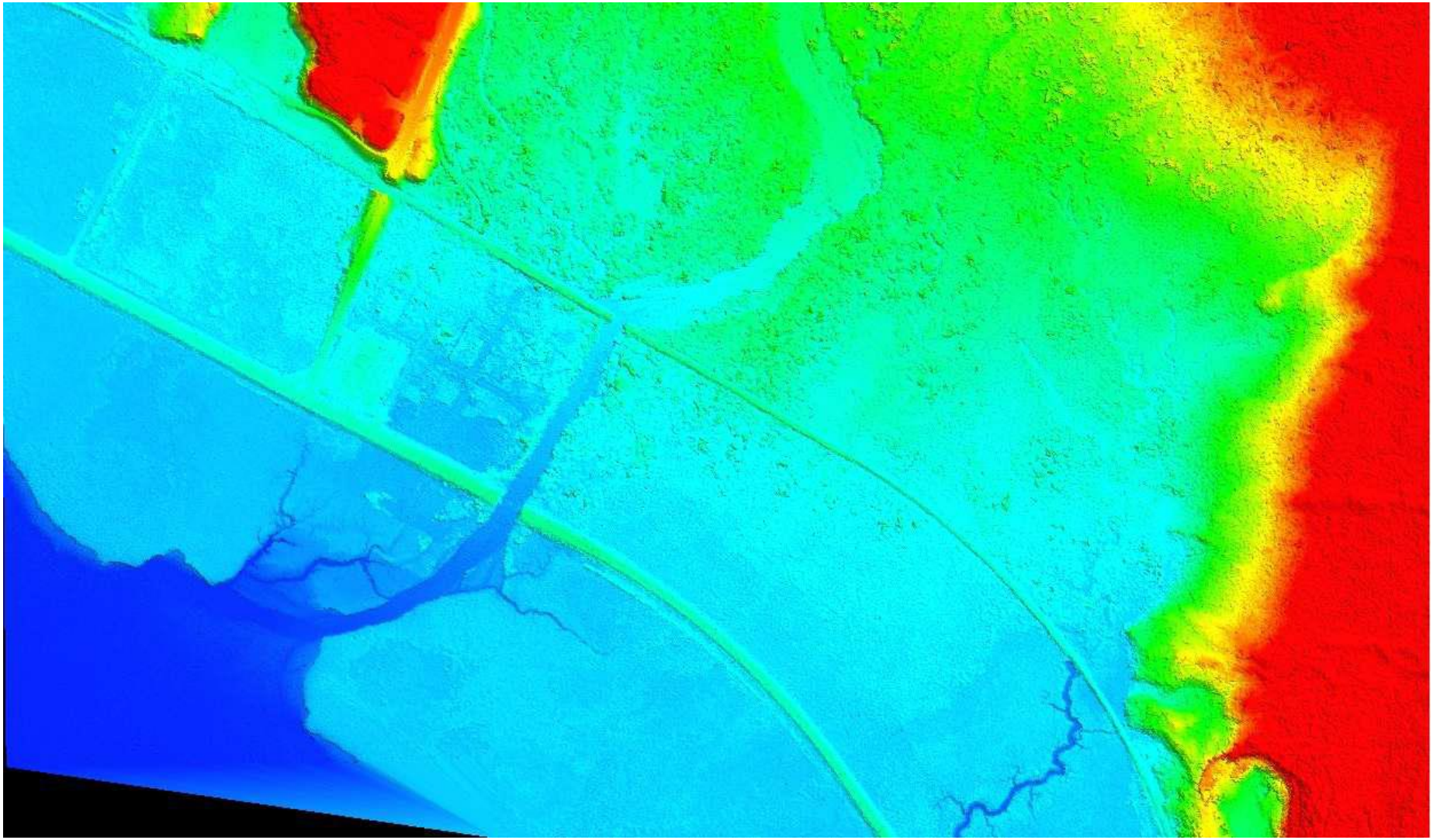
DSM



DEM

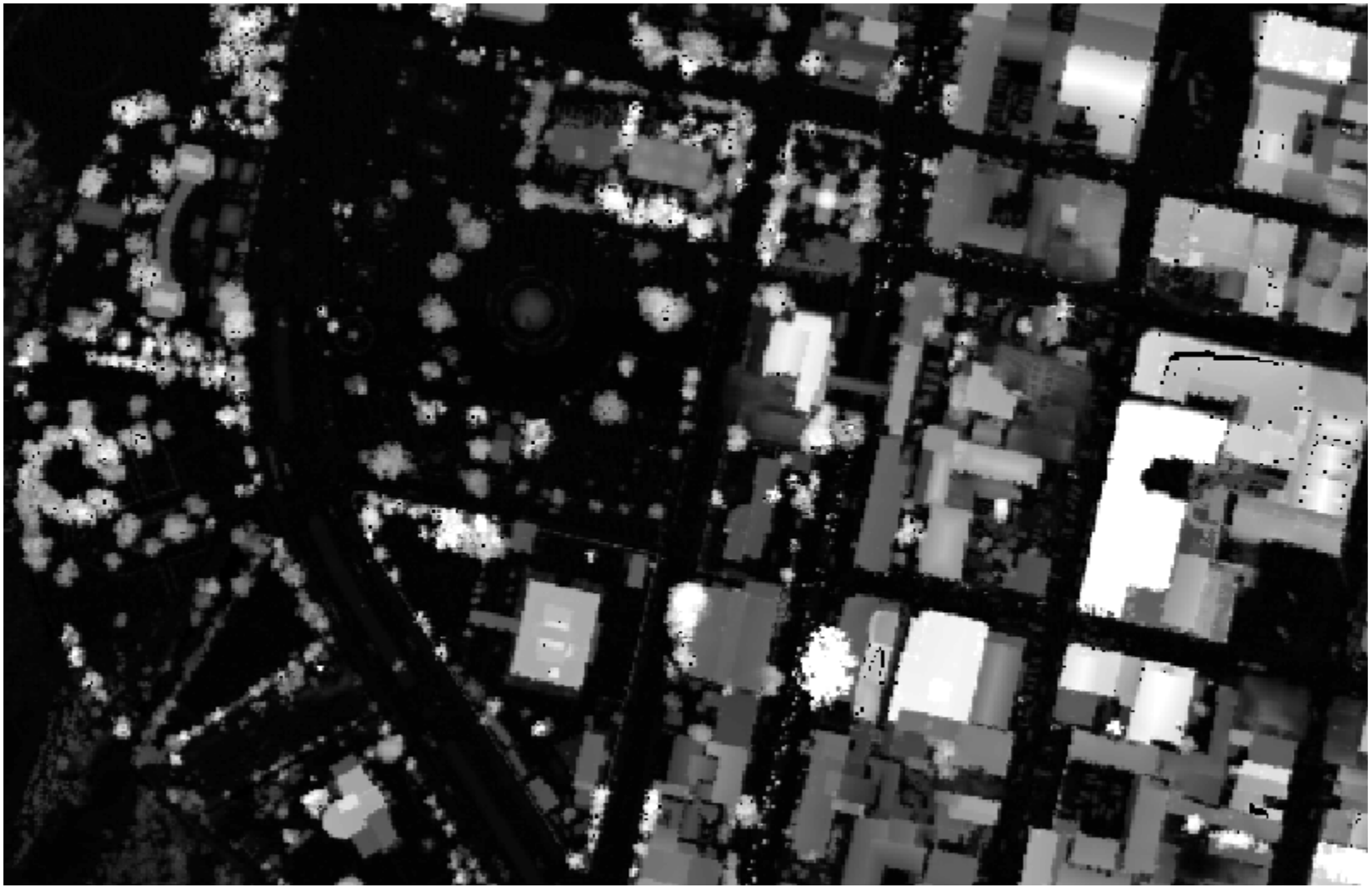


Digital Elevation Model (DEM)



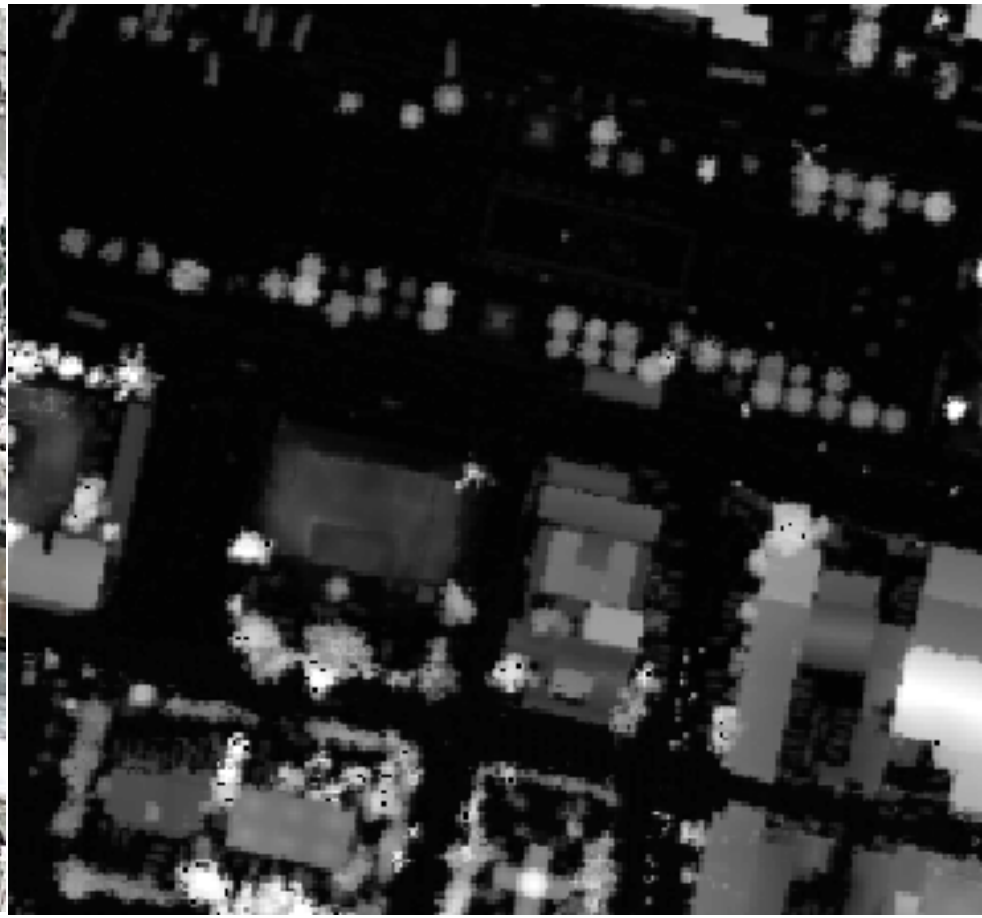
Courtesy: Charlie Barnwell Michael Baker Jr. Corp.

LiDAR nDSM



GeoEye-1 and LiDAR nDSM





DSM and DEM

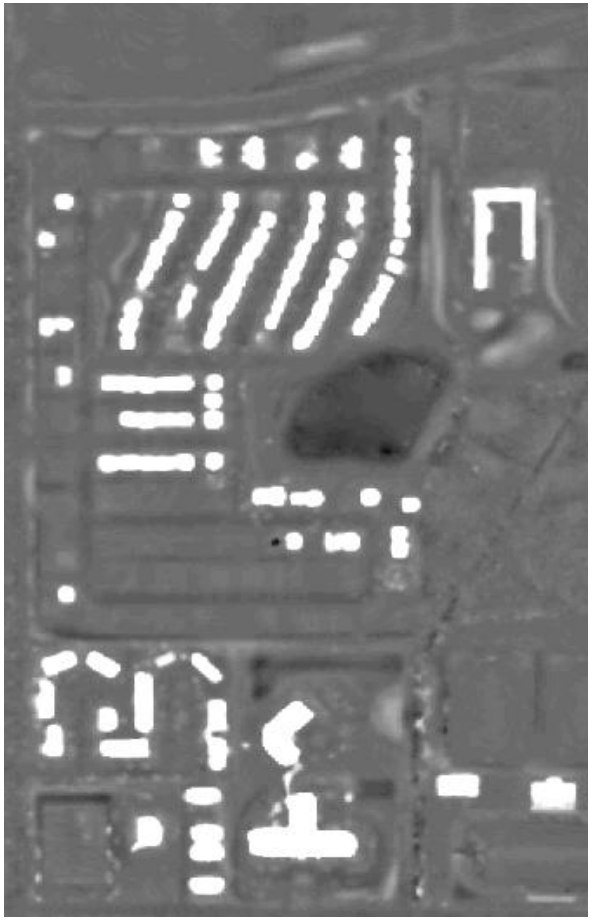

Value
High : 827.284
Low : 505.848
1000 m


Value
High : 733.846
Low : 506.832
1000 m

Ortho Image and nDSM

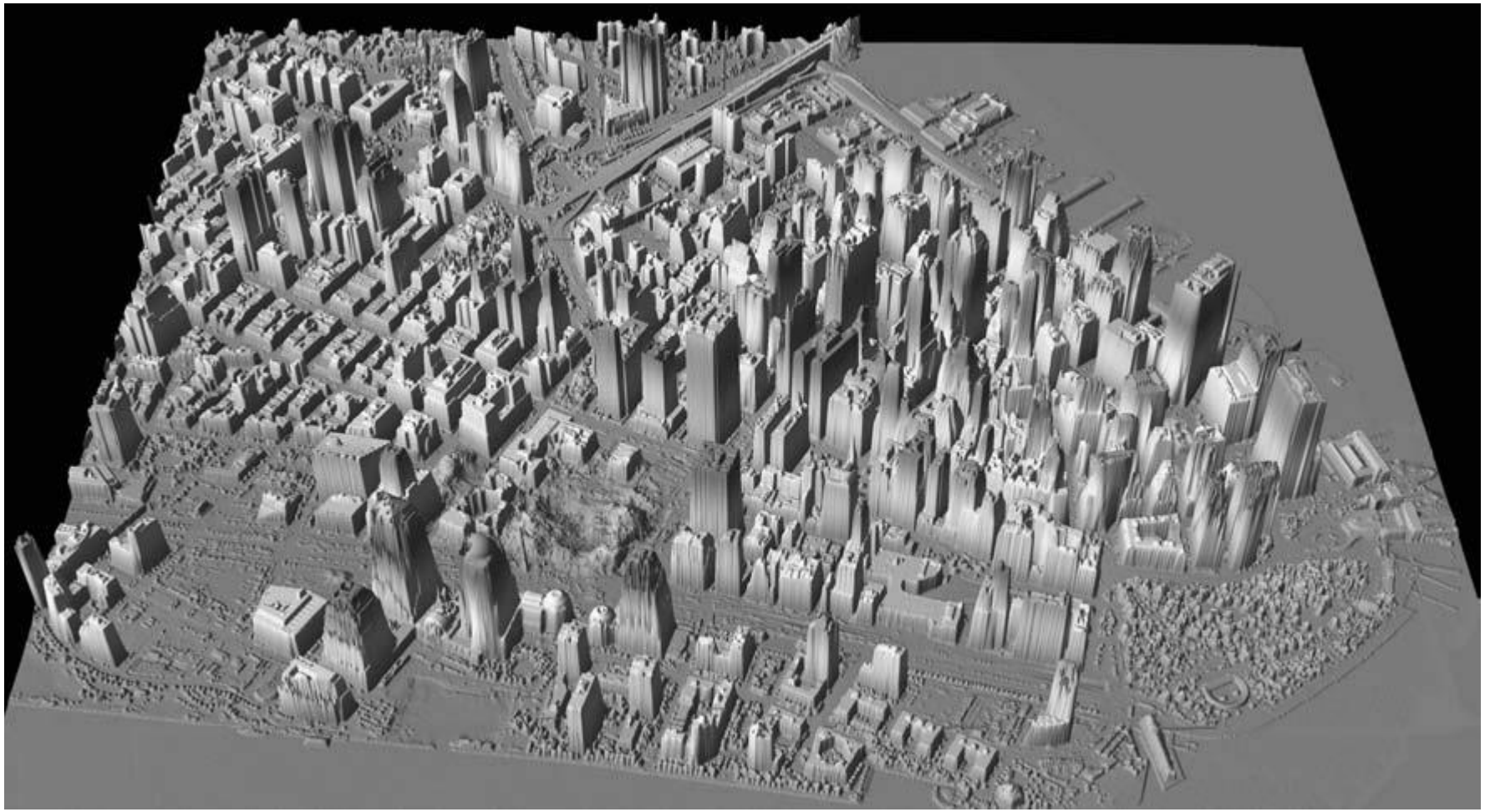


2005 Image



nDSM

LiDAR DATA



3D Urban Visualization

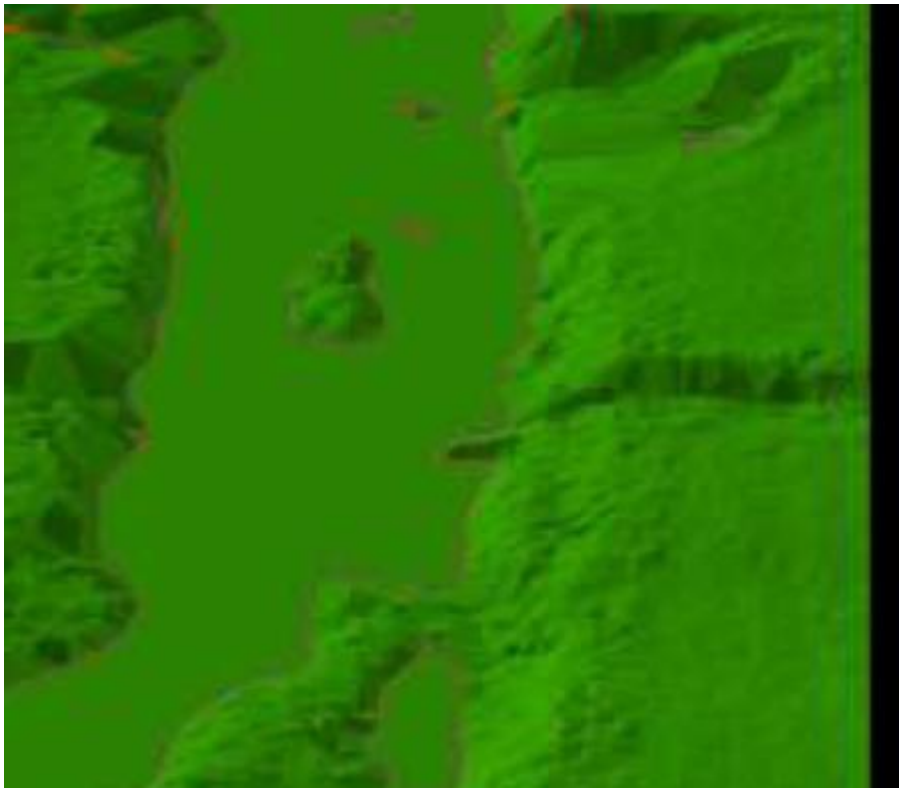
Digital Elevation Model

- Reliability with which the Earth surface can be represented is determined by horizontal spacing and vertical accuracy of the elevation
- Sudden elevation change (such as cliffs or across drainage features) cannot be represented well by gridded DEMs
- Interpolation on either sides is misleading, since the slopes at such points are not continuous
- Some gridded DEM includes *break lines* which indicates the discontinuities in the topographic surface
- A break line is a line of terrain where there is a significant change in slope. Break lines fall on ridge lines, valleys, ditches and other natural or artificial features where there is a change in slope
- When interpolating elevations from DEM in the vicinity of break line, only points on the same side of the break line are used

Use of Breaklines



Aerial Image



DEM

Courtesy: Charlie Barnwell Michael Baker Jr. Corp.

Advantages & Applications

- Quick data delivery (DEM and DSM) with in 24 hours of data acquisition
- Can operate at night
- Convenience and flexibility in flight planning
- LiDAR data are used for
  - Forested area coverage for tree heights and density
  - Detailed information concerning building densities, urban structures, and building footprints
  - Highway planning
  - Pipeline route planning
  - Designing wireless communication systems in urban regions
  - Floodplain mapping

Signal-to-Noise ratio

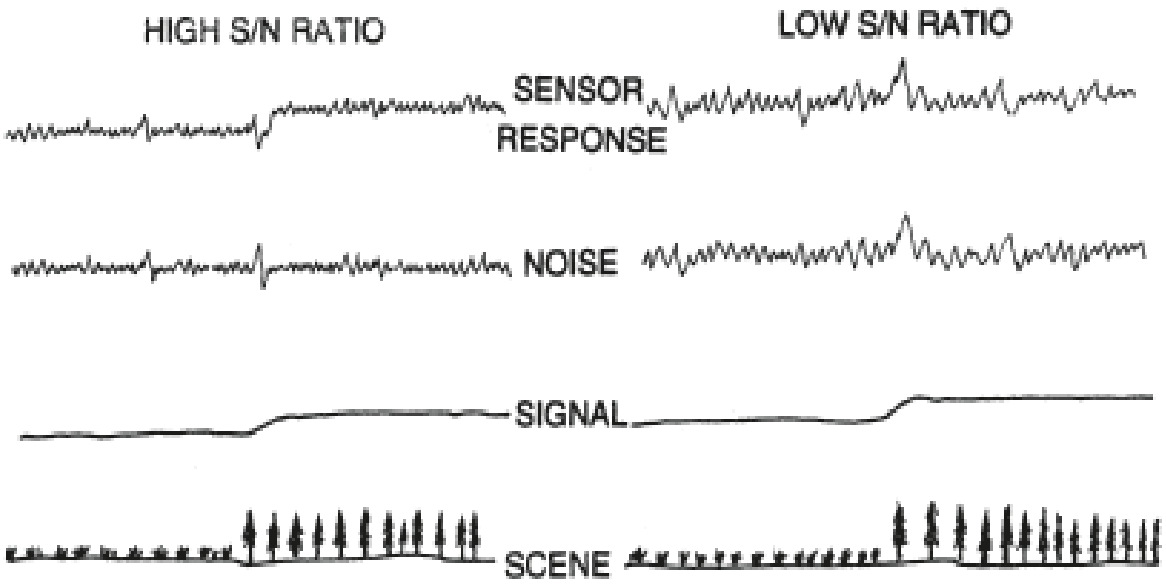


FIGURE 4.7. Signal-to-noise (S/N) ratio. At the bottom, a hypothetical scene is composed of two cover types. The signal records this region, with only a small difference in brightness between the two classes. Atmospheric effects, sensor error, and other factors contribute to noise, which is added to the signal. The sensor then records a combination of signal and noise. When noise is small relative to the signal (left: high S/N ratio), the sensor conveys the difference between the two regions. When the signal is small relative to noise (right, low S/N ratio) the sensor cannot portray the difference in brightness between the two regions.

Campbell:pp 106-107